Drug tolerance takes shape: persistent homology reveals cyclic cell-state plasticity in EGFR-mutant lung cancer

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Abstract

Drug-tolerant persister cells drive relapse after EGFR-targeted therapy, yet no scRNA-seq method quantifies the reversibly cyclic state transitions that produce them: standard trajectory tools expose no scalar measure of how robustly cells trace a closed loop in state-space. We present an open-source pipeline that applies persistent homology and Mapper to longitudinal scRNA-seq with permutation-null testing, cell-cycle controls, and gene attribution. Across five EGFR-mutant lung-cancer datasets ($\sim$31,000 analysed cells), maximum H_1 persistence rose monotonically with osimertinib exposure: PC9 1.41 $\rightarrow$ 3.78 (2.7$\times$, $p < 0.01$), PDX YU-006 2.81 $\rightarrow$ 6.08 (2.2$\times$), YU-003 2.79 $\rightarrow$ 3.85 (1.4$\times$); in a 14-patient EGFR-mutant cohort (Maynard 2020) pooled H_1 grew 5.37 $\rightarrow$ 7.13 $\rightarrow$ 8.05 across treatment-naive, persister, and progressive disease, with 3/3 matched pre/post biopsies increasing. The signal survives Tirosh ablation, stringent S/G2M regression, and Harmony batch correction. Loop-associated transcripts include EMT programs in cell line and PDX, with significant system-specific overlap. Max H_1 is a reproducible cross-system phenotype for cell-state plasticity; code is MIT-licensed.

Keywords: topological data analysis; persistent homology; Mapper; single-cell RNA-seq; drug tolerance; EGFR-mutant lung cancer; cell-state plasticity; computational framework.

1 Background

Trajectory methods systematically miss cyclic cell-state structure

Single-cell RNA sequencing has transformed our view of cellular heterogeneity, but the dominant trajectory toolkit for ordering cells — Monocle and Slingshot (tree backbones), PAGA (connectivity graph), and RNA velocity (vector field) — was designed to summarise progression rather than to quantify reversible cycling [1]. PAGA admits non-tree graphs and velocity defines a field, but neither yields a scalar that scores how robustly cells trace a closed loop across analysis scales. That gap matters whenever cell populations reversibly transition between states — as in the epithelial-to-mesenchymal transition (EMT), the cell cycle, and adaptive drug tolerance [2, 3] — because a reversible cycle is mathematically a one-dimensional hole in the data's point cloud.

Topological data analysis as a framework-level remedy

An intuitive picture for biomedical readers. Topology studies the properties of shapes that survive continuous deformation — a coffee mug and a donut are the same shape because each has exactly one hole, and one can be smoothly deformed into the other (fig. 1a). Plot each cell as a point in gene-expression space: cells transitioning linearly between states trace an elongated cloud, while cells that reversibly cycle (e.g. epithelial → intermediate → mesenchymal → epithelial) arrange into a closed *1-cycle* — a loop that is not the boundary of any filled-in disc within the data (fig. 1b). Tree-based trajectory methods force such a loop into a branching tree and discard the cyclic information. Persistent homology measures how robustly a 1-cycle survives as the connecting distance scale grows: the lifespan of the longest bar in the resulting *barcode* (fig. 1c) is the *maximum H_1 persistence* used throughout this work. Mapper complements this with an interpretable graph in which nodes group transcriptionally similar cells and edges connect related nodes [4].

Formally, persistent homology returns a multiset of birth–death pairs (the persistence diagram) over a filtered sequence of simplicial complexes [5], and Mapper [4] returns a complementary graph summary. Both have been applied individually to bulk cancer transcriptomes [6] and developmental scRNA-seq [7]; prior work has not integrated them into a reproducible, open-source framework for longitudinal scRNA-seq with formal null-model testing and standard controls — the gap our contribution addresses.

Drug-tolerant persisters as a testbed

Drug-tolerant persister cells (DTPs) in EGFR-mutant non-small cell lung cancer (NSCLC) provide an ideal testbed for a topology-based framework. First-line osimertinib gives a median progression-free survival of roughly 18 months, yet essentially every patient relapses from residual disease, and no approved drug specifically targets the persister state [8, 9]. DTPs are a small, slow-growing subpopulation that survives targeted therapy not by acquiring mutations but by entering a reversible, quiescent transcriptional state — behaving like cells hibernating through drug pressure and waking up when conditions relax [3, 10–12]. Multiple public scRNA-seq datasets cover longitudinal treatment with erlotinib (1st- generation TKI) or osimertinib (3rd-generation TKI) at cell-line and patient-derived xenograft (PDX) scales, and lineage tracing [3] has established that DTPs contain cycling subpopulations — but the topological signature of this dynamic has not been formally quantified.

Contribution

We present an open-source pipeline that applies persistent homology (H_0, H_1) and the Mapper algorithm to longitudinal scRNA-seq, with integrated permutation-null testing, two independent cell-cycle controls, Wasserstein diagram comparison, and Mapper-based gene attribution. We validate it on five independent EGFR-mutant lung-cancer datasets covering two TKIs (erlotinib, osimertinib), three biological scales (cell line, PDX, patient tumour), and a 14-patient longitudinal cohort. The maximum H_1 persistence rises monotonically with drug exposure across the validation cohorts, survives both cell-cycle controls, and is biologically anchored by EMT-program enrichment in loop-associated transcripts (cell-line and PDX overlap $p = 1.25 \times 10^{-11}$). All analyses are reproducible from raw GEO data via a single Makefile target under MIT license.

2 Results

Erlotinib-treated PC9 cells display robust H_1 persistence (max $H_1 = 3.78$ at day 9)

We first asked whether drug-treated PC9 cells (harboring EGFR exon 19 deletion E746-A750del) exhibit non-trivial H_1 features consistent with cyclic state-space structure. We analyzed GSE134839 [12], a Drop-seq time-series of PC9 cells treated with 2 µM erlotinib at days 0, 1, 2, 4, 9, and 11 (fig. 2). After standard QC and preprocessing (Methods), 2,942 cells across six timepoints were projected into the top-30 principal components and used to compute the persistence diagrams via the Vietoris–Rips filtration.

All drug-treated timepoints showed non-zero H_1 persistence, with maximum values ranging from 1.08 (D11) to 1.77 (D9) (fig. 3, table 1). Persistence barcodes showed an increased number of long H_1 bars at intermediate and late timepoints. Pairwise Wasserstein distances between persistence diagrams revealed that D0 is separated from all drug-treated timepoints in persistence-diagram space (Wasserstein distance $W = 213$–265 between D0 and treated timepoints vs. $W = 26$–66 among treated timepoints — roughly an order of magnitude larger), indicating that the distribution of topological features differs most strongly between untreated and treated conditions (fig. 3b).

Per-timepoint sample sizes (200–379 cells) left individual permutation tests underpowered (closest D9 $p = 0.098$, D2 $p = 0.126$, near the peak of persister emergence reported by Aissa et al.); the larger validation cohorts below recover $p < 0.01$ (table 2).

Loops persist after Tirosh cell-cycle ablation across five cohorts (ratio 0.71–1.11)

To rule out cell cycle oscillation as a confounder, we removed 39 of 97 Tirosh et al. cell-cycle genes [13] present in our HVG set, re-scaled, and re-computed PCA and persistent homology. The ratio of H_1 after to before cell-cycle ablation ranged from 0.71 to 1.11 across all six timepoints (mean 0.90 ± 0.14), with D11 actually *increasing* after cycle-gene removal (ratio = 1.11) — indicating that the observed cyclic topology is not driven by cell-cycle genes (fig. 4, table 3). This ratio-based test is the standard control in the field and has been interpreted as evidence against cell-cycle artifacts.

(Per-timepoint nulls on the ablated discovery cohort were $p = 0.11$–0.58, expected at the small n further degraded by removing 39 HVGs; larger validation cohorts retain $p < 0.01$; Table S1.)

Stringent S/G2M score regression preserves loops in all five cohorts (ratio 1.07–4.05)

Stringent `sc.pp.regress_out(S_score, G2M_score)` on 1,000-cell subsamples preserves loops across five cohorts (post/pre ratio 1.07–4.05, mean 1.97, every value > 1; table 4; full discussion in Supplementary Methods S13). The osimertinib cohort additionally contains zero Tirosh genes in its HVG set, ruling out Tirosh-specific contribution to that cohort by absence.

Osimertinib drives a 2.7× monotonic max H_1 increase over 14 days

On GSE150949 [3], a PC9 lineage-tracing osimertinib time-series (56,419 cells subsampled to 5,000; D0/D3/D7/D14; 4,999 post-QC), max H_1 rose monotonically 1.41 → 1.82 → 2.67 → 3.78 (2.7×; fig. 5), with permutation tests confirming $p < 0.01$ at D7 and D14 (table 2).

PDX xenografts replicate the osimertinib-driven cycle (YU-006: 2.2×, YU-003: 1.4×)

In GSE243562 [14] — two EGFR-mutant PDX models (YU-003, YU-006) under matched untreated and osimertinib-residual conditions, 18,874 cells subsampled to 1,500 per group — both models showed an increase in max H_1 under residual disease: YU-003 2.79 → 3.85 (1.4×, $p < 0.01$); YU-006 2.81 → 6.08 (2.2×, $p < 0.01$) (fig. 6, table 2). YU-006 matches the cell-line D14 effect; YU-003 is more modest but directionally consistent. This replicates the cell-line finding in an *in vivo* system with stromal, vascular, and host-immune context.

Max H_1 in drug-treated cells approaches patient-tumor levels

To establish a topological reference for untreated human lung adenocarcinoma, we analysed GSE131907 [15] — a 208,506-cell LUAD atlas across 44 patients — subsampling 5,000 primary-tumour epithelial cells (tLung), then 1,500 for VR computation. Treatment-naive patient LUAD shows max $H_1 = 4.77$, substantially higher than treatment-naive PC9 cell line (1.41) or PDX (2.79, 2.81) baselines (fig. 7). The Kim atlas is mixed-driver (EGFR-, KRAS- and other-mutant tumours; not genotype-selected), so this is a generic LUAD reference rather than an EGFR-specific baseline.

Drug treatment moves cell-line and PDX topology from the cell-line-like regime ($H_1 \approx 1$–3) toward the mixed-driver patient range ($H_1 \approx 4$–5; fig. 7, closest at osi D14 and YU-006 residual) — a qualitative cross-system comparison confounded by platform and cell count; the load-bearing claim is the within-system monotonic increase (fig. 5, fig. 6).

Timepoint-label permutation null confirms drug-specific signal

The gene-label null tests for non-random topology but not for drug specificity, so we ran a timepoint-label null on osimertinib D14 vs. D0 (drug-exposure annotation shuffled across the pooled 2,499 cells; H_1 recomputed in each permutation). Over 50 permutations, the observed $\Delta H_1 = 2.07$ exceeded the 95th percentile (1.18) in 0/50 trials, giving $p < 0.02$ (Phipson–Smyth +1 correction) — direct evidence that H_1 depends on drug exposure.

EMT genes dominate loop-associated transcripts in both PC9 and PDX systems

What do the H_1 loops represent biologically? To identify the transcriptional programs underlying loop formation, we constructed Mapper graphs on the analyzed cell line data using EMT score as filter function (Methods). The Mapper graph contained 20 nodes and 16 edges; gene-level connectivity scores quantify how strongly each gene varies between Mapper nodes versus globally.

In the cell-line analysis, top-connected genes were canonical EMT markers (*TPM1*, *KRT8*, *KRT19*); MSigDB Hallmark EMT ranked first in top-100 pathway enrichment. The PDX analysis recovered a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*) — 3/33 vs. 8/33 Hallmark EMT genes in the cell-line top 100 — consistent with an EMT-like plastic state realised through a partially shared repertoire. Eleven genes overlap between the two top-100 lists (including *KRT19*, *LGALS3*); hypergeometric $p = 1.25 \times 10^{-11}$ against the 13,955-gene background of features expressed in both analyses, $p = 2.1 \times 10^{-4}$ against the more conservative HVG-intersection background (3,000 genes). Differential expression recovered EMT-related signatures in both systems with system-specific second-order pathways (cholesterol homeostasis downregulated in the cell line; estrogen-response upregulated in PDX, a pathway known to interact with EMT via estrogen-receptor–mediated transcription in NSCLC [16]). Full gene-level lists are in Supplementary Tables S2, S2′, and S4.

Robustness of gene attribution. Two sensitivity checks support the EMT readout. With PC1 as an unbiased Mapper filter, only 4/33 Hallmark EMT genes appear in the top 100 (vs. 8/33 with the EMT filter) and cell-cycle markers (*MKI67*, *TOP2A*, *HMGB2*) co-rank with EMT markers, so the loops reflect a multi-program state-space rather than a pure EMT signature; PC1↔EMT top-100 overlap is 53/100. Variance-normalised connectivity (dividing each gene's contribution by its global SD) raises Hallmark-EMT hits to 13/33 in the top 100 and drops housekeeping genes out, arguing the enrichment is not an expression-magnitude artefact. Full tables: Supplementary S2 (EMT filter), S2′ (PC1 filter), S9 (variance-normalised).

The framework is modular, reproducible, and generalises beyond drug tolerance

The pipeline is fifteen independent modules coordinated by a single `make pipeline` target (Methods; Supplementary Methods S8). The same framework ran without modification on Drop-seq and 10x cell-line data (6,508–56,419 raw cells), 10x PDX data with automated mouse-gene filtering (22,032 cells), and a 208,506-cell patient atlas. Applying it to a new dataset needs only (i) a QC'd and normalised `AnnData` object with a sample/timepoint column, (ii) a Mapper filter (PC1, EMT score, or diffusion component 1 are shipped as defaults), and (iii) a permutation-test subsampling budget (1,000 cells per group is sufficient at $p < 0.05$). We anticipate the framework will generalise to other scRNA-seq perturbation contexts including chemotherapy, immunotherapy escape, and stress-induced dedifferentiation. Resampling stability (CV 8.9–23.5% with ordering preserved), replication under alternative persistence-diagram summaries (total persistence and persistence entropy), and Mapper-parameter robustness (mean Spearman $\rho = 0.994$ across a 4×5 parameter sweep) are reported in Supplementary Methods S10–S12.

The monotonic trend survives Harmony batch correction

To rule out that the observed H_1 growth reflects batch or platform effects rather than intrinsic biology, we applied Harmony batch correction [17] to the top-50 PCA components with dataset-appropriate batch keys (full parameters in Supplementary Methods S7). Max H_1 was then recomputed on the Harmony-corrected embeddings; Figure 8 summarises the comparison.

The osimertinib monotonic trend survives Harmony correction ($1.35 \rightarrow 3.94$; $2.9\times$, vs. $1.71 \rightarrow 3.78$ raw); the YU-006 untreated→residual difference is preserved ($2.59 \rightarrow 5.19$ Harmony vs. $2.36 \rightarrow 5.01$ raw); YU-003 becomes more modest but directionally consistent; the Kim LUAD naive baseline is robust (Harmony 4.52 vs. raw 4.77). Full values: Supplementary Table S10. The H_1 growth across drug exposure therefore reflects biology rather than within-dataset batch structure.

In 14 EGFR-mutant NSCLC patients, max H_1 grew monotonically TN → PER → PD ($5.37 \rightarrow 7.13 \rightarrow 8.05$)

To directly test the framework on clinical tissue, we analysed the full EGFR-mutant subset of Maynard et al. [18]: 13,244 Smart-seq2 cells across 14 patients with biopsies at three treatment stages — treatment-naive (TN, 3,287 cells), persister/residual disease (PER, 4,893 cells), and progressive disease (PD, 5,064 cells). Across 1,500-cell subsamples per stage, pooled max H_1

grew monotonically 5.37 → 7.13 → 8.05 (TN-to-PD 1.50×; fig. 9a).

Per-patient matched trajectories. Three patients had matched TN + post-treatment biopsies: TH218 (1.22 → 1.65, 1.35×), TH226 (2.11 → 6.86, 3.25×; → 5.01 PD, 2.37×), and AZ003 (5.32 → 8.50, 1.60×; fig. 9b). Post-treatment max H_1 exceeded TN in 3/3 matched pairs (median 1.60×, range 1.35–3.25×; one-sided sign test $p = 0.125$; underpowered at $n = 3$ but directionally consistent; Supplementary Table S11). Raw max H_1 spans $\sim$ 1.2–7 across patients at any stage — inter-patient biological and technical heterogeneity — so the within-patient trajectory (always up) is a stronger signal than absolute magnitude, supporting use of max H_1 as a *relative-change* readout.

This patient-cohort result is the most clinically proximal evidence we provide: the same topological signature that rises under drug in PC9 and PDX also rises in real EGFR-mutant NSCLC patient biopsies, in a hypothesis-generating three-of-three matched design.

3 Discussion

A reproducible framework rather than a single dataset finding

The primary contribution of this work is an open-source, modular framework for applying topological data analysis to longitudinal scRNA-seq data, with integrated permutation null testing, cell-cycle ablation, Wasserstein diagram comparison, and Mapper-based gene attribution. The five datasets analyzed here are a deliberately heterogeneous benchmark (two drugs, three biological scales, two PDX models, one patient atlas, one 14-patient longitudinal cohort) chosen to stress-test reproducibility rather than to maximize statistical significance on any single cohort. This reframes the question from "does drug tolerance produce topological cyclicity?" to "is topological cyclicity a reproducible, quantitative phenotype across biological systems that can be detected and compared with a standardized pipeline?" Our answer is yes: max H_1 increases monotonically with osimertinib exposure in three independent systems (PC9 cell line, YU-003 PDX, YU-006 PDX), and all seven 1,000-cell permutation tests in table 2 reach $p < 0.05$ (6/7 at $p < 0.01$; the lone exception is the PC9 osimertinib baseline D0 at $p = 0.030$).

Convergent topology, divergent gene programs

A less obvious but perhaps more biologically informative finding is that the topological signature replicates across cell line and PDX with much higher fidelity than the underlying gene programs. Cell line top-connected genes are dominated by canonical EMT markers (*TPM1*, *KRT8*, *KRT19*), while PDX top-connected genes are a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*, *AGR2*). The 11-gene overlap between the two top-100 lists, although highly significant statistically (hypergeometric $p = 1.25 \times 10^{-11}$), is a minority of either list. We interpret this as evidence that the cell-state plasticity underlying DTP emergence is realized through partially different but related gene expression repertoires depending on microenvironment — an observation consistent with recent findings on context-dependent EMT programs [19]. The topological statistic abstracts over these specific gene choices, making it a more stable cross-system phenotype than any individual gene signature.

Relationship to existing literature

Our work is complementary to and independent of Oren et al. [3], who used Watermelon lineage tracing to establish that PC9 DTPs contain cycling subpopulations, and Aissa et al. [12], who identified discrete DTP transcriptional states in the source GSE134839 dataset. Our framework provides an orthogonal method of detecting cyclic behavior directly from the scRNA-seq transcriptome without lineage-tracing infrastructure, and extends the signal to PDX data Oren et al. did not analyze. Relative to Rizvi et al. [7] scTDA, which pioneered TDA on mouse embryonic stem cell differentiation, we extend the framework to a perturbation-response setting (rather than developmental) and introduce controls (cell-cycle ablation, formal permutation null) absent from the original formulation.

The broader non-genetic drug-tolerance literature — YAP-mediated senescent DTPs [20], melanoma minimal-residual-disease four-state plasticity [21], and the Marine et al. review of non-genetic resistance [22, 23] — provides the biological context. Our contribution against this backdrop is a quantitative, reproducible statistic (topologically invariant in loop existence, metric-dependent in numerical scale; Supplementary Methods S4) for detecting and comparing such structures.

Limitations

Platform and batch confounding. Cross-system comparisons mix Drop-seq (GSE134839) with 10x Chromium (others); the cross-system ordering (cell line < PDX < patient) is reported only qualitatively. Within-system comparisons are platform-matched and survive Harmony batch correction (fig. 8), ruling out within-dataset batch structure as the source. A full scVI cross-platform integration [24] is deferred.

Permutation null specification. The gene-label null tests only for non-random point-cloud structure; the complementary timepoint-label null on osi D14 vs. D0 ($p < 0.02$; Results) directly tests drug dependence.

Patient-cohort validation. The three Maynard matched pre/post pairs (sign-test $p = 0.125$; Results) are hypothesis-generating, not confirmatory; the pooled 14-patient TN→PER→PD monotonic increase is supportive but observational. A prospective paired-biopsy study would be needed to power the within-patient comparison.

Cell-cycle ablation stringency. Tirosh ablation removes 39/97 cell-cycle genes (standard but not exhaustive — quiescence and SASP markers are not in the Tirosh list); the more demanding S/G2M-score regression (ratio 1.07–4.05 across five cohorts) and the PC1-filter Mapper (cell-cycle genes co-rank with EMT markers) together indicate the topology is multi-program rather than purely EMT, and is not a cell-cycle artefact.

Implications and outlook

If drug tolerance operates through topologically cyclic plasticity, state-trapping combinations that collapse the cycle may outperform single-state-targeting strategies. The cross-scale reproducibility of the signature argues that max H_1 is a quantitative phenotype for cell-state plasticity, suitable for benchmarking drug responses across labs and platforms and likely to generalise to adjacent perturbation-response settings.

Why might cyclic plasticity be adaptive? We speculate that a robust 1-cycle corresponds to a Waddington-style limit cycle around a shallow basin [25, 26], making cyclic plasticity adaptive under fluctuating selection as bet-hedging [27, 28]; distinguishing drug-induced from drug-selected cycles would require lineage tracing. Adaptive therapy [29, 30] and state-trapping combinations become testable hypotheses against this readout.

4 Conclusions

Drug treatment does not simply push lung-cancer cells along a one-way road to resistance: it reshapes the cell-state landscape into a form that admits reversible cycling — including EMT-like programs — consistent with the cycling subpopulations established by lineage tracing in PC9 [3]. The maximum H_1 persistence introduced here quantifies how robust that cycling becomes, rises monotonically with osimertinib across cell-line and PDX validation cohorts, survives two independent cell-cycle controls and Harmony batch correction, and tracks loop-associated EMT programs in both systems. We expect the readout to generalise to other reversible-plasticity settings where standard trajectory methods fall short.

5 Methods

Datasets

GSE134839 [12]: PC9 cells treated with 2 µM erlotinib; six Drop-seq DGE matrices (D0, D1, D2, D4, D9, D11); 6,508 total cells. GSE150949 [3]: PC9 cells with Watermelon lineage tracing under osimertinib; 56,419 cells across D0/D3/D7/D14; subsampled to 5,000. GSE243562 [14]: two EGFR-mutant PDX models (YU-003, YU-006) under osimertinib; 22,032 cells across four samples (untreated and residual disease for each model). GSE131907 [15]: treatment-naive patient LUAD atlas; 208,506 cells across 44 patients; subsampled to 5,000 tLung epithelial cells, then a further 1,500 cells for tractable Vietoris–Rips computation (filter and seed details in Supplementary Methods S1).

Preprocessing

We used scanpy 1.11 [31] with standard QC (`min_genes` = 200, `max_genes` = 5000, `max_pct_mito` = 20%, `min_cells` = 3), total-count normalisation to 10,000 counts per cell, log1p transformation, 3,000 highly-variable genes, regression of total counts and mitochondrial percentage, scaling to max 10, and PCA with 50 components (`random_state` = 42). HVG batch keys per cohort: `timepoint` for GSE134839 and GSE150949, `pdx` for GSE243562, and `Sample` for GSE131907 (rationale in Supplementary Methods S2). For PDX data, mouse genes were removed by a symbol-pattern heuristic (Supplementary Methods S3; 29% of features filtered; limitations in Discussion). Cell-cycle scores were computed from Tirosh et al. [13] S- and G2/M-phase gene lists via `sc.tl.score_genes_cell_cycle`.

Persistent homology

We used `ripser` 0.6 [32, 33] to compute the Vietoris–Rips filtration up to H_1 with the Euclidean metric on the top-30 PCA coordinates; homology is computed over $\mathbb{F}_2$ coefficients (ripser default). Maximum persistence is defined as $\max(\text{death} - \text{birth})$ over finite H_1 bars; this is a hybrid statistic whose existence is a topological invariant but whose numerical scale is metric-dependent (Supplementary Methods S4). Wasserstein distances between diagrams were computed with `persim` 0.3 using the 2-Wasserstein metric. The p-Wasserstein distance between persistence diagrams is stable under bounded perturbations of the input point cloud given a finite-degree-p total-persistence hypothesis [34, 35].

Principal component count sensitivity

Max H_1 depends on the number of PCs retained: across $n_{\text{PCs}} \in \{10, 20, 30, 50\}$ absolute values vary within a factor of two, but the drug-treated > untreated ordering is preserved at every PC count. The $n_{\text{PCs}} = 30$ default follows prior scTDA work [7]. Full sweep numbers and the rationale for the hybrid-statistic framing are in Supplementary Methods S4–S5 and Supplementary Table S5.

Permutation tests

For each group, we shuffled expression values within each cell (permuting gene labels independently for every cell), re-fit PCA using `sklearn`, and recomputed H_1 max persistence. The p-value is the fraction of null permutations with max $H_1 \geq$ the observed value. Discovery cohort used 500 permutations per timepoint; validation cohorts used 100 permutations on 1,000-cell subsamples for tractability.

Mapper

We used `kmapper` 2.0 [36] with an auto-tuned DBSCAN clusterer (`eps` set to the 50th percentile of 10th-nearest-neighbor distances in each cover element). Default parameters were $n_{\text{cubes}} = 15$

and perc_overlap = 0.3. Parameter sensitivity was assessed across $n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$ and overlap $\in \{0.2, 0.3, 0.4, 0.5\}$.

EMT score (used as Mapper filter for the cell-line analysis) was computed via `scanpy.tl.score_genes` on a 33-gene MSigDB Hallmark EMT cassette (28 of 33 present in GSE134839; full gene list in Supplementary Methods S6).

Gene connectivity

For each gene g and Mapper node N with cells C_N, we computed $|\text{mean}(g \mid C_N) - \text{mean}(g \mid \text{all cells})| \times |C_N|$, summed over all nodes, and divided by total cells. Higher scores indicate genes that vary more between Mapper nodes than globally.

Pathway enrichment

Gene set enrichment was performed with `gseapy 1.1` against MSigDB_Hallmark_2020 and GO_Biological_Process_2021.

Computational environment

All analyses ran on Python 3.11 in a conda environment specified in `envs/environment.yml`; random seed 42 is used for every stochastic operation. Full package versions and platform notes (macOS arm64 and Linux x86_64): Supplementary Methods S14.

Pipeline architecture and unit tests

The framework is organised as fifteen numbered modules (`01_preprocess.py` through `15_maynard_full_cohort` that each accept a standard `AnnData` object and produce typed intermediate outputs. Users can substitute their own datasets by providing an `AnnData` file with a `timepoint` observation column. Full module list and the `pytest` unit-test suite (8 tests, all passing) are described in Supplementary Methods S8–S9.

Code availability

Code is available at https://github.com/htlin222/sctda-cancer-plasticity (MIT license). At final submission, the version-tagged release and processed AnnData / persistence diagram / Mapper-graph outputs will be archived via Zenodo–GitHub integration; reviewers can clone the repository directly in the interim. The complete pipeline is reproducible via `make pipeline` from raw GEO data to all figures and tables. A unified `sctda` CLI is planned for v2.0; the current pipeline is invoked via the Makefile and individual `scripts/0X_*.py` files. A `CITATION.cff` file at the repository root provides machine-readable citation metadata for software-citation tooling.

Data availability

All raw scRNA-seq data used in this study are publicly available from the Gene Expression Omnibus under accessions GSE134839, GSE150949, GSE243562, and GSE131907; the Maynard et al. 2020 EGFR-mutant cohort was obtained from the authors' public Google Drive archive referenced in Maynard et al. [18]. Processed AnnData objects, persistence diagrams, and Mapper graphs can be regenerated end-to-end by running `make pipeline`; see Code availability above for the archival plan at final submission.

Declarations

Ethics approval and consent to participate

Not applicable — all analyses use previously published, de-identified public datasets.

Consent for publication

Not applicable.

Availability of data and materials

All raw data and processed intermediate files are available as described in the Methods section (Data availability). Supplementary tables accompanying this manuscript: S1 (cell-cycle ablation permutation p-values), S2 (EMT-filter Mapper gene connectivity), S2′ (PC1-filter Mapper gene connectivity), S3 (Mapper parameter sweep), S4 (PDX gene attribution), S5 (Max H_1 vs principal component count), S6 (functional persistence summaries: max H_1, total persistence, persistence entropy across ten cohorts), S7 (subsample variance: mean $\pm$ SD of max H_1 across five random seeds for four cohorts), S8 (Mapper parameter-sweep Spearman rank correlations), S9 (variance-normalized Mapper gene connectivity), S10 (Harmony batch-correction raw vs. corrected max H_1 comparison), S11 (Maynard 2020 full EGFR-mutant cohort per-patient max H_1 and within-patient trajectory statistics (14 patients, 3 matched pre/post pairs)), S-stringent-CC (stringent cell-cycle regression ratios; reproduced as Table 4).

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

H.-T.L. (first author) and Y.-H.T. (corresponding author) jointly conceived the study and designed the analytical strategy. H.-T.L. implemented the pipeline, performed all analyses, produced all figures, and wrote the original draft (CRediT: Conceptualisation, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualisation). Y.-H.T. supervised the project, contributed methodological and validation-strategy review, and revised the manuscript (CRediT: Conceptualisation, Methodology, Supervision, Writing – review & editing, Project administration). Both authors read and approved the final manuscript.

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Figures

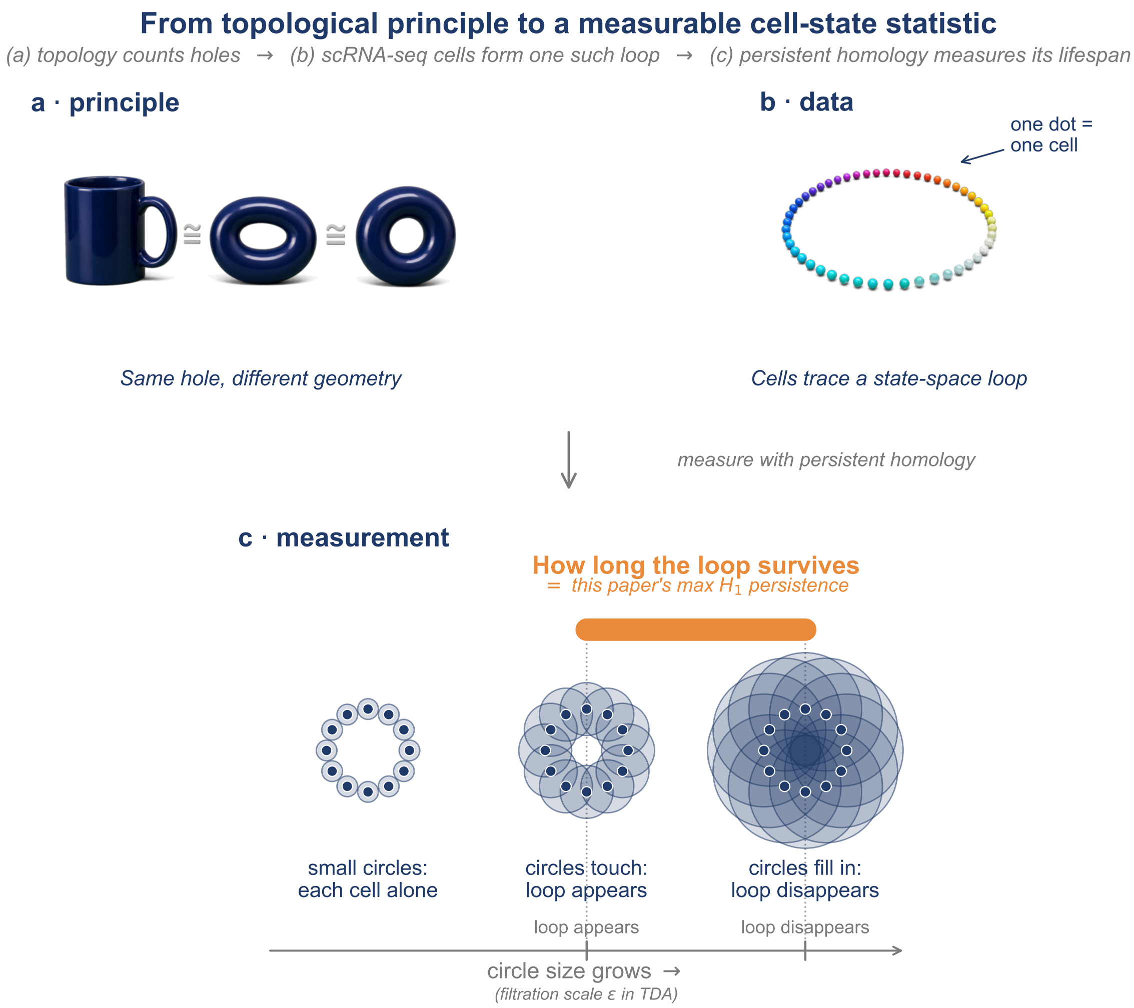


Figure 1: **Topology distinguishes shapes that geometry cannot.** (a) Mug and donut share one hole and are continuously deformable into one another — topologically equivalent despite different geometry. (b) Cells reversibly cycling among states lie on a closed loop in state-space; the loop itself is the invariant. (c) Persistent homology summarises this as a barcode: short bars reflect noise (H_0 components); one long H_1 bar (orange) marks a robust loop. Its lifetime is the *maximum* H_1 *persistence* used throughout this work.

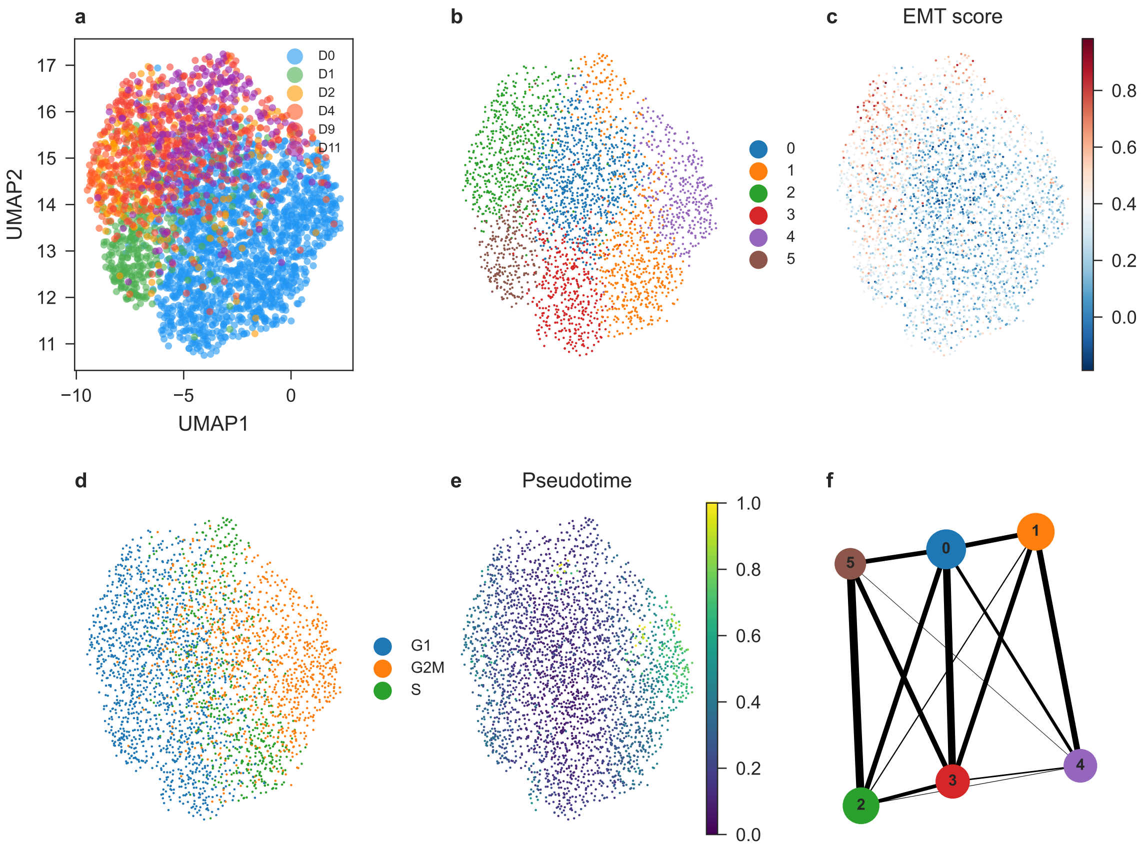
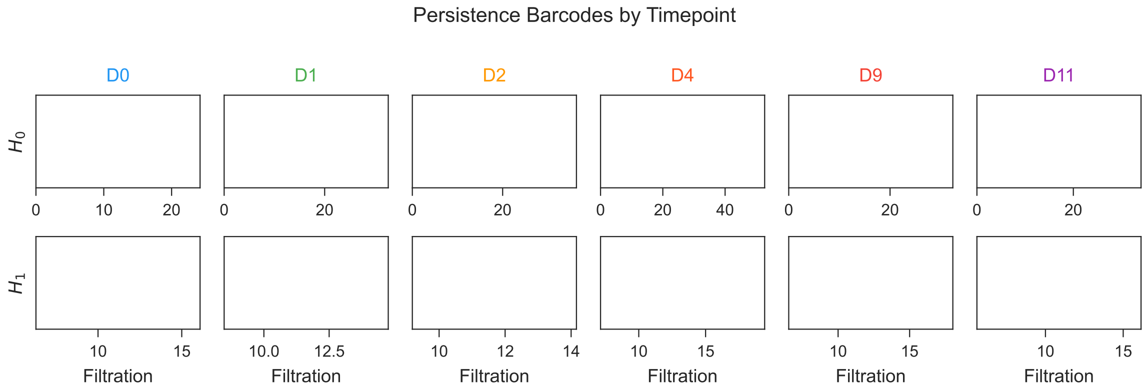
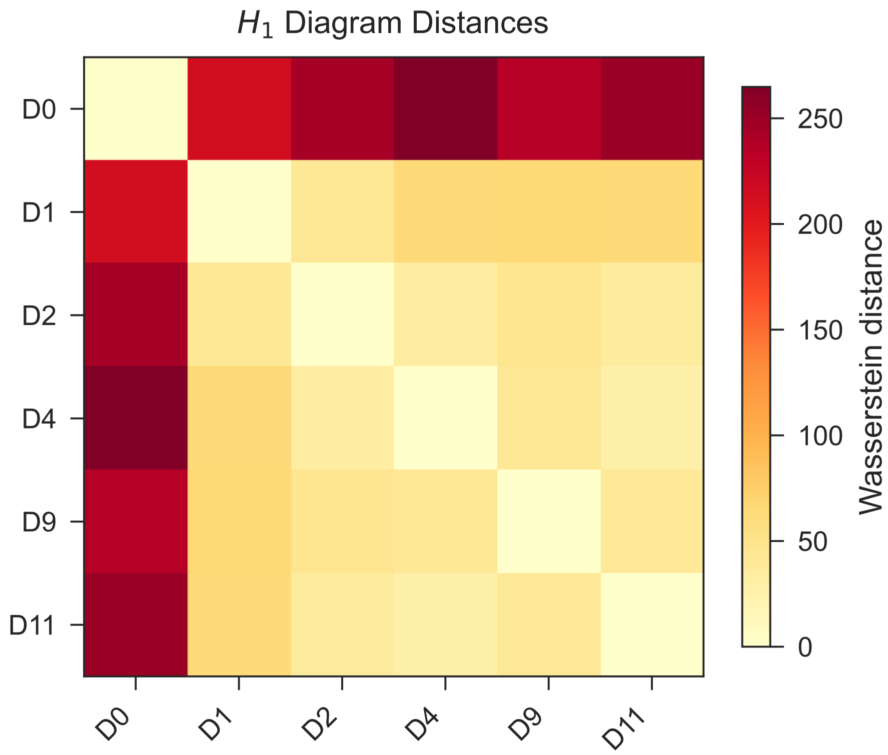


Figure 2: **Study design and cell-line baseline analysis.** (a) UMAP of PC9 erlotinib time course colored by timepoint. (b) Leiden clusters. (c) EMT score. (d) Cell cycle phase. (e) Diffusion pseudotime. (f) PAGA graph. $n = 2{,}942$ cells across six timepoints.

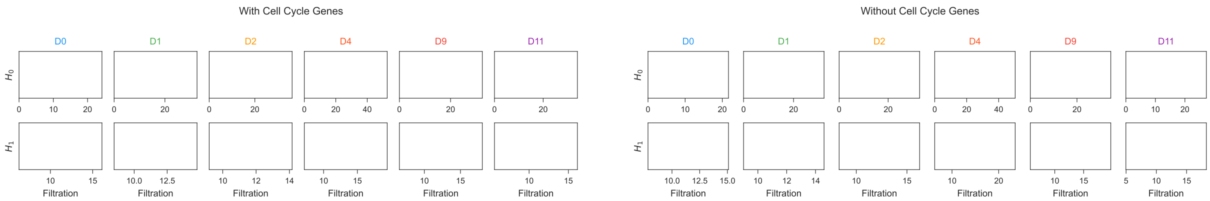


(a) Persistence barcodes by timepoint (top: H_0; bottom: H_1).



(b) Pairwise Wasserstein distance heatmap between per-timepoint H_1 persistence diagrams. D0 is separated from drug-treated timepoints in persistence-diagram space.

Figure 3: **Topological analysis of PC9 erlotinib time course.** (a) Persistence barcodes by timepoint. Permutation tests' closest approaches to significance were at D9 ($p = 0.098$) and D2 ($p = 0.126$); neither reaches conventional $p < 0.05$. (b) Wasserstein distance heatmap between per-timepoint H_1 persistence diagrams.



(a) With cell-cycle genes.

(b) Without cell-cycle genes.

Figure 4: **Cell cycle ablation.** (a,b) Persistence barcodes with and without cell-cycle genes. Ratio > 0.7 at every timepoint indicates loops are not cell-cycle artifacts.

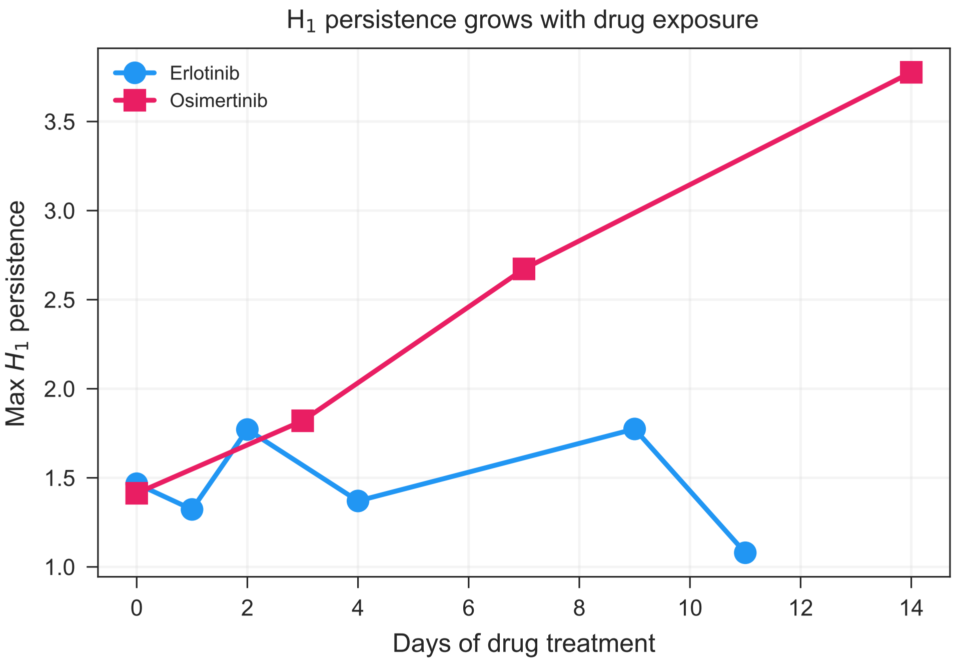


Figure 5: **Osimertinib validation in PC9.** Max H_1 vs. days for both erlotinib and osimertinib time courses. Osimertinib shows a 2.7× monotonic increase from D0 to D14.

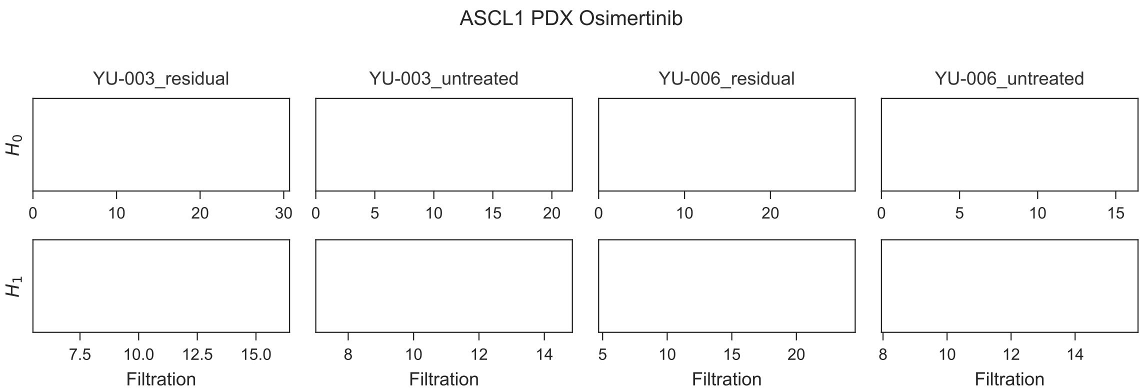


Figure 6: **Osimertinib validation in patient-derived xenografts.** Persistence barcodes for YU-003 and YU-006 under untreated and residual conditions. Both models show increased H_1 persistence under residual disease.

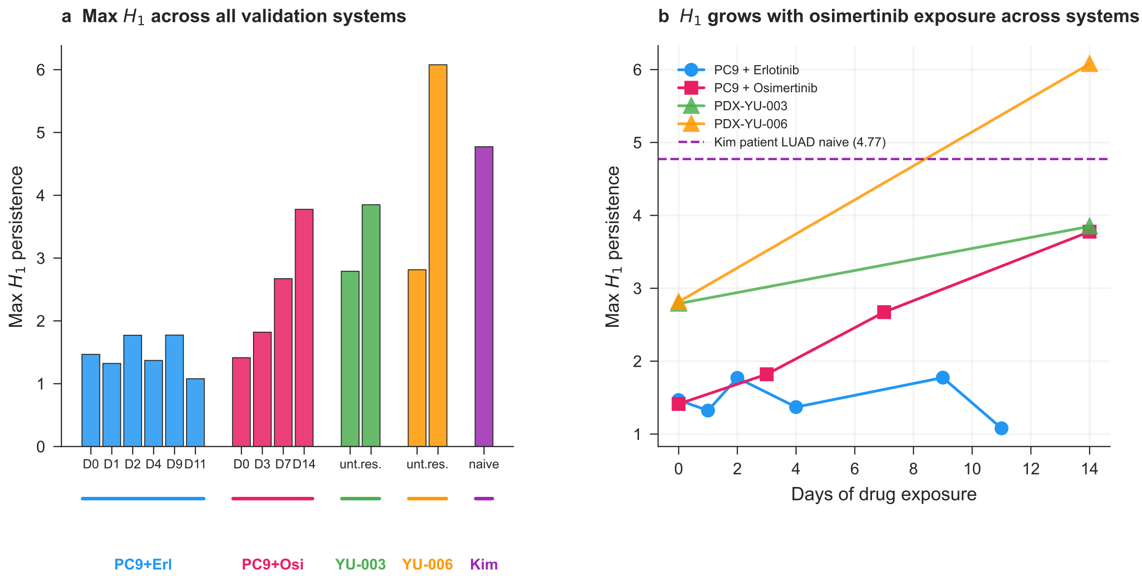


Figure 7: **Master comparison across all biological systems.** (a) Max H_1 bar chart across all conditions. (b) Max H_1 vs. days for all drug-treated systems, with patient-naive baseline as a reference line. Drug treatment pushes cell line and PDX topology toward patient-tumor-like complexity.

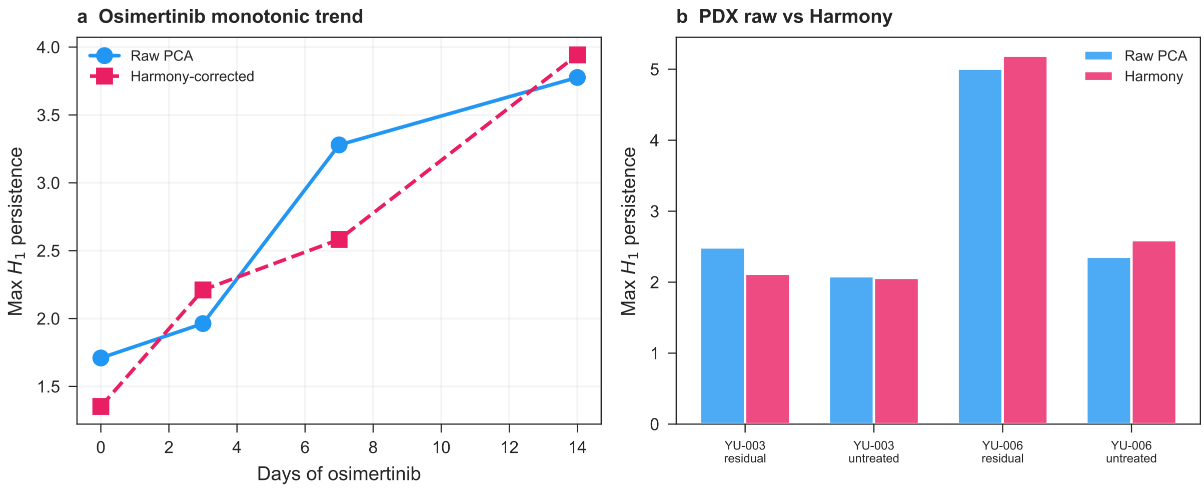


Figure 8: **Harmony batch correction sensitivity.** (a) Max H_1 vs. days of osimertinib exposure in the PC9 cell line cohort (GSE150949), computed on raw PCA (blue) and on Harmony-corrected embeddings (pink, dashed). Both show monotonic growth from D0 to D14. (b) Max H_1 per PDX group (untreated vs. residual disease) in GSE243562, raw (blue) vs. Harmony-corrected (pink). The YU-006 pre/post effect is preserved after batch correction.

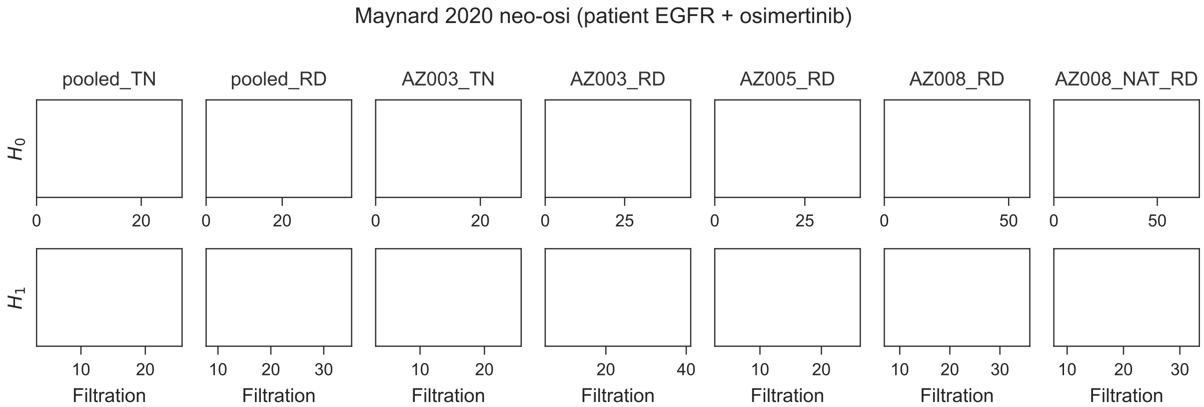


Figure 9: **Patient-cohort validation in the Maynard 2020 EGFR-mutant NSCLC cohort.** Persistence barcodes (H_0 top, H_1 bottom) for the Maynard et al. 2020 Smart-seq2 dataset, EGFR-mutant subset (14 patients, 13,244 cells). Full cohort pooled monotonic growth: TN H_1 = 5.37 → PER 7.13 → PD 8.05. In 3 of 3 patients with matched pre/post biopsies (TH218, TH226, AZ003), max H_1 increased post-treatment (median 1.60×, range 1.35–3.25×). See Supplementary Table S11 for full per-patient trajectories.

Tables

Table 1: **Per-timepoint H_1 summary (GSE134839, erlotinib).** Permutation p-values from 500 within-cell gene-label shuffles.

Timepoint	n cells	Max H_1	# H_1 features	Perm. p (500)
D0	1560	1.47	1393	0.394
D1	324	1.32	255	0.552
D2	230	1.77	124	0.126
D4	200	1.37	54	0.144
D9	379	1.77	204	0.098
D11	249	1.08	124	0.266

Table 2: **Validation cohort H_1 permutation tests (100 permutations, 1,000-cell subsamples).** Values from 1,500-cell subsample used in main analysis; permutation tests used 1,000-cell subsamples (see Methods). † YU-003 residual value is from the 1,500-cell main analysis (consistent with Figure 6); permutation tests at 1,000-cell subsamples show elevated variance (CV 23.5%, Supplementary Table S7) with this specific subsample at max $H_1 = 2.07$ — we report the main analysis value here for consistency with other entries in this table. ‡ Per-cohort max H_1 values vary across subsample sizes (1,000 vs. 1,500 cells) due to the strong subsample dependence of persistent homology on cell count (Supplementary Table S7, CV 8.9–23.5%); values reported in Table 2 use the 1,000-cell permutation-test subsample for internal consistency with the p-value column. Note that PC9 osimertinib D0 max H_1 is reported as 1.41 in the main analysis (5,000-cell subsample) and 1.74 here (1,000-cell permutation subsample); these are different subsamples of the same underlying data and are not inconsistent.

Dataset	Group	Observed H_1	p-value
PC9 + Osimertinib	D0	1.74	0.030
PC9 + Osimertinib	D7	3.08	< 0.01
PC9 + Osimertinib	D14	3.78	< 0.01
PDX YU-006	Untreated	3.62	< 0.01
PDX YU-006	Residual	6.08	< 0.01
PDX YU-003	Untreated	2.79	< 0.01
PDX YU-003	Residual	3.85	$< 0.01^\dagger$

Table 3: **Cell cycle ablation H_1 summary.** Ratio of max H_1 after cell-cycle gene removal to before. Ratios ≥ 0.71 across all timepoints indicate the cyclic structure is not a cell cycle artifact.

Timepoint	H_1 with CC	H_1 without CC	Ratio	Verdict
D0	1.47	1.30	0.88	Persists
D1	1.32	1.15	0.87	Persists
D2	1.77	1.57	0.89	Persists
D4	1.37	0.97	0.71	Persists
D9	1.77	1.39	0.78	Persists
D11	1.08	1.20	1.11	Persists

Table 4: **Stringent cell-cycle regression (`sc.pp.regress_out` on S and G2/M scores).** Ratio of post-regression to pre-regression max H_1 across five cohorts. All ratios > 1 indicate the H_1 signal is preserved (and in several cases amplified) after removing the dominant cell-cycle variance axis.

Group	n cells	H_1 original	H_1 regressOut	Ratio
Erl D0	1000	1.27	2.85	2.25
Erl D9	379	1.77	1.89	1.07
Erl D11	249	1.08	4.37	4.05
PDX YU-006 untreated	1000	2.89	3.67	1.27
PDX YU-006 residual	1000	4.65	5.96	1.28

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